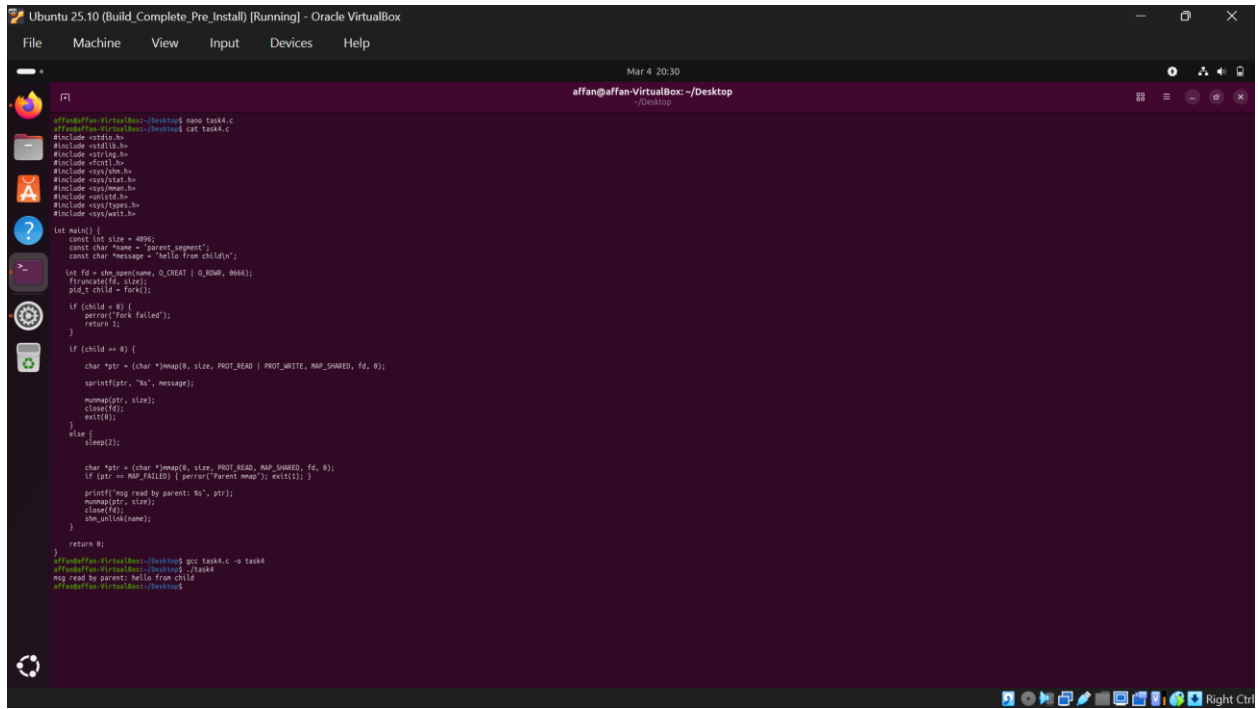


Task 3:



```
affan@affan-VirtualBox: ~/Desktop
affan@affan-VirtualBox:~/Desktop$ nano task4.c
affan@affan-VirtualBox:~/Desktop$ cat task4.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/jm.h>
#include <sys/stat.h>
#include <sys/mman.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <sys/time.h>

int main() {
    const int size = 4096;
    const char *name = "parent segment";
    const char *message = "hello from child\n";

    int fd = shm_open(name, O_CREAT | O_RDWR, 0666);
    ftruncate(fd, size);
    pid_t child = fork();

    if (child == 0) {
        perror("fork failed");
        return 1;
    }

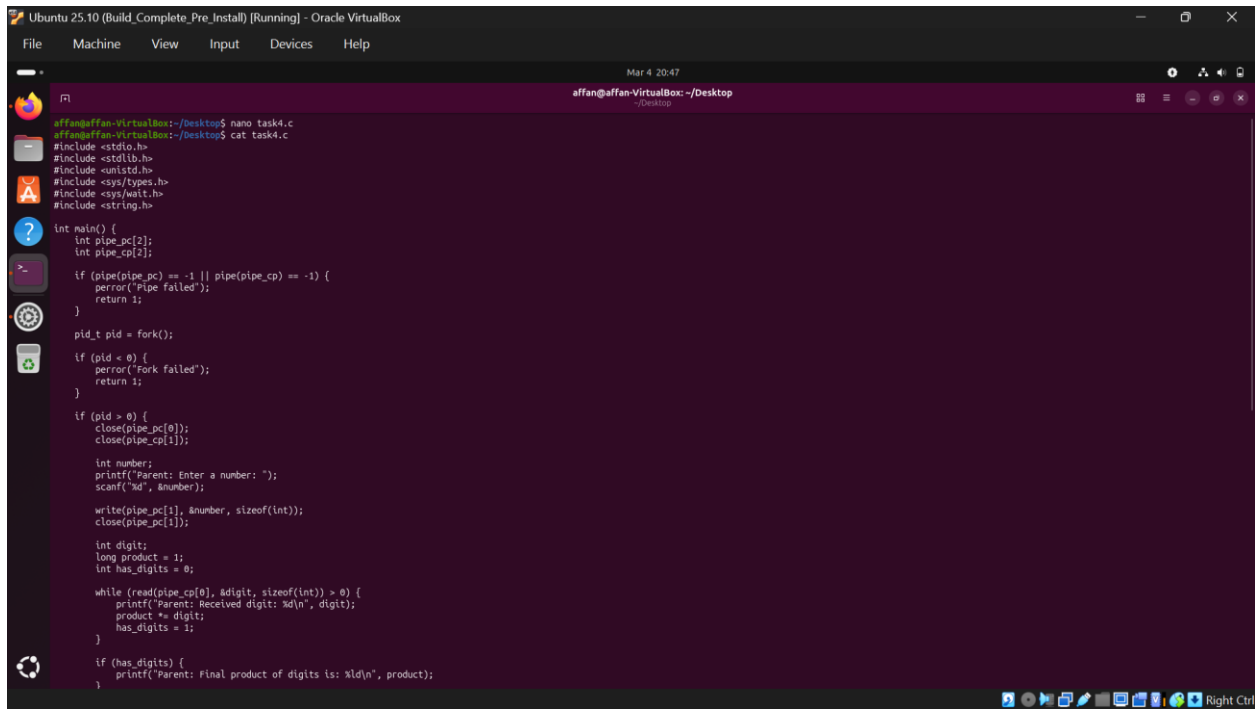
    if (child == 0) {
        char *ptr = (char *)mmap(0, size, PROT_READ | PROT_WRITE, MAP_SHARED, fd, 0);
        sprintf(ptr, "%s", message);
        munmap(ptr, size);
        close(fd);
        exit(0);
    } else {
        sleep(2);

        char *ptr = (char *)mmap(0, size, PROT_READ, MAP_SHARED, fd, 0);
        if (ptr == MAP_FAILED) { perror("parent mmap"); exit(1); }
        printf("msg read by parent: %s", ptr);
        munmap(ptr, size);
        close(fd);
        shm_unlink(name);
    }

    return 0;
}

affan@affan-VirtualBox:~/Desktop$ gcc task4.c -o task4
affan@affan-VirtualBox:~/Desktop$ ./task4
msg read by parent: hello from child
affan@affan-VirtualBox:~/Desktop$
```

Task 4:



```
affan@affan-VirtualBox: ~/Desktop
affan@affan-VirtualBox:~/Desktop$ nano task4.c
affan@affan-VirtualBox:~/Desktop$ cat task4.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {
    int pipe_cp[2];
    int pipe_pc[2];

    if (pipe(pipe_cp) == -1 || pipe(pipe_pc) == -1) {
        perror("Pipe failed");
        return 1;
    }

    pid_t pid = fork();

    if (pid == 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        close(pipe_cp[0]);
        close(pipe_cp[1]);

        int number;
        printf("Parent: Enter a number: ");
        scanf("%d", &number);
        write(pipe_cp[1], &number, sizeof(int));
        close(pipe_cp[1]);

        int digit;
        long product = 1;
        int has_digits = 0;

        while (read(pipe_cp[0], &digit, sizeof(int)) > 0) {
            printf("Parent: Received digit: %d\n", digit);
            product *= digit;
            has_digits = 1;
        }

        if (has_digits) {
            printf("Parent: Final product of digits is: %ld\n", product);
        }
    }
}
```

```
Ubuntu 25.10 (Build_Complete_Pre_Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Mar 4 20:47
affan@affan-VirtualBox: ~/Desktop
~/Desktop

if (has_digits) {
    printf("Parent: Final product of digits is: %ld\n", product);
}

close(pipe_cp[0]);
wait(NULL);
}
else {
    close(pipe_pc[1]);
    close(pipe_cp[0]);

    int received_num;
    read(pipe_cp[0], &received_num, sizeof(int));
    close(pipe_cp[0]);

    if (received_num < 0) received_num = -received_num;

    if (received_num == 0) {
        int zero = 0;
        write(pipe_cp[1], &zero, sizeof(int));
    } else {
        int digits[15], count = 0;
        while ((received_num > 0)) {
            digits[count++] = received_num % 10;
            received_num /= 10;
        }
        for (int i = count - 1; i >= 0; i--) {
            write(pipe_cp[1], &digits[i], sizeof(int));
        }
    }

    close(pipe_cp[1]);
    exit(0);
}

return 0;
}

affan@affan-VirtualBox:~/Desktop$ gcc task4.c -o task4
affan@affan-VirtualBox:~/Desktop$ ./task4
Parent: Enter a number: 12345
Parent: Received digit: 1
Parent: Received digit: 2
Parent: Received digit: 3
Parent: Received digit: 4
Parent: Received digit: 5
Parent: Final product of digits is: 120
affan@affan-VirtualBox:~/Desktop$
```